



CARBELIM SCIENTIFIC & TECHNICAL TEAM

CARBELIM

Product Portfolio Performance Report

**Comparative Analysis of Energy Consumption, CO₂
Sequestration,
and Oxygen Generation**

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1. Executive Summary

This report presents a comprehensive performance overview of Carbelim's complete photobioreactor (PBR) product line, comprising seven models spanning culture volumes from 2 L (Miniforest) to 500 L (PRO15). The analysis covers tank specifications, annual electricity consumption and operating cost, biomass productivity, CO₂ biological fixation, oxygen evolution, and equivalence to mature oak tree carbon sequestration.

All projections of biomass, CO₂, and O₂ output are derived from a validated reference dataset obtained on the Carbelim 60 L photobioreactor using calibrated analytical methods. Biomass productivity ($1.8 \text{ g L}^{-1} \text{ day}^{-1}$) was determined by gravimetric dry weight analysis of the culture, in which a known culture volume was harvested, dried to constant weight, and the resulting dry biomass mass normalised to culture volume and cultivation time. The carbon fraction of this dry biomass was independently quantified using a CHNS elemental analyser, which determines the percentage carbon content by mass through combustion analysis. CO₂ fixation was then calculated by converting the measured carbon mass to its CO₂-equivalent using the molecular weight ratio of CO₂ to carbon (44:12), yielding 197.64 g CO₂ fixed per day (72.14 kg per year) on the 60 L reference system. Oxygen evolution was derived from the fixed CO₂ using the stoichiometric mass ratio of oxygenic photosynthesis (32:44), yielding 143.74 g O₂ per day (52.47 kg per year).

This experimentally derived productivity rate is applied uniformly across the product range, since all units share the same core microalgal culture system and differ primarily in culture volume and form factor. Across the seven products, annual CO₂ fixation ranges from 2.17 kg (Miniforest) to 596.22 kg (PRO15), while energy costs range from ₹700.8 to ₹22,425 per year. The larger-volume units demonstrate substantially better cost efficiency per kilogram of CO₂ sequestered, with the Park Bench (Large) and PRO15 offering the lowest cost per kg CO₂ in the lineup.

2. Product Range Overview

Carbelim's product line is engineered around a shared microalgal photobioreactor platform, scaled to serve use cases ranging from compact personal desktop units to large-scale outdoor installations. Table 1 summarizes the tank volume, annual energy consumption, and operating cost for each product.

Table 1: Product Specifications, Energy Consumption & Operating Cost

Product	Tank Volume	Energy (kWh/yr)	Cost/Year (₹)	Daily Energy (kWh)
Miniforest	2 L	87.60	700.80	0.24
Air Forest Indoor	70 L	876	7008.00	2.40
Air Forest Outdoor	100 L	1109.60	8876.80	3.04
PRO15	500 L	2803.20	22425.60	7.68
Wall Mount	30 L	233.60	1868.80	0.64
Park Bench (Large)	500 L	2336.00	18688.00	6.40
Park Bench (Small)	150 L	876.00	7008.00	2.40

The Miniforest is the smallest unit at 2 L, designed for personal desktop use, consuming just 87.60 kWh/year at a cost of ₹700.80/year. At the other end of the range, the PRO15 (500 L) is Carbelim's largest installation, consuming 2803.20 kWh/year at 22425.60/year — reflecting its substantially greater air-processing capacity. Notably, the Air Forest Indoor (70 L) and Park Bench (Small) (150 L) share identical annual energy consumption of 876 kWh/year, despite their differing volumes, indicating differences in component configuration or duty cycle between these two models.

3. Biomass Productivity

Biomass output for each product is calculated by applying the validated productivity rate of 1.8 g L⁻¹ day⁻¹ — established from the Carbelim 60 L reference system — to each unit's culture volume:

$$\text{Daily Biomass (g)} = \text{Tank Volume (L)} \times 1.8$$

This linear scaling reflects the fact that biomass growth rate per litre of culture is a function of the consistent photosynthetic and nutrient-delivery conditions maintained across all Carbelim systems, independent of tank size. Table 2 presents the resulting daily and annual biomass production for each product.

Table 2: Daily and Annual Biomass Production

Product	Daily Biomass (g)	Annual Biomass (kg)	Tank Volume
Miniforest	3.24	1.18	2 L
Air Forest Indoor	115.20	42.05	70 L
Air Forest Outdoor	168.30	61.43	100 L
PRO15	891.00	325.21	500 L
Wall Mount	49.50	18.07	30 L
Park Bench (Large)	819.00	298.94	500 L
Park Bench (Small)	253.80	92.64	150 L

Biomass output scales directly with tank volume: the PRO15 produces 891 g of dry biomass per day (325.21 kg/year), roughly 275 times the output of the Miniforest's 3.24 g/day (1.18 kg/year). The Park Bench (Large) (819 g/day) closely trails the PRO15, reflecting its near-comparable 500 L volume.

4. CO₂ Fixation & O₂ Evolution

4.1 Methodology

Biomass productivity for the reference 60 L system was established through gravimetric dry weight analysis: a known volume of culture was harvested at defined time intervals, the biomass separated by filtration or centrifugation, and the resulting solids dried to constant weight in an oven. Normalising the dry biomass mass to culture volume and elapsed cultivation time yielded the productivity figure of 1.8 g L⁻¹ day⁻¹.

The carbon content of this dry biomass was determined directly using a CHNS elemental analyser, which combusts a known mass of sample and quantifies the percentage of carbon, hydrogen, nitrogen, and sulphur by mass through combustion-based elemental analysis. This measured carbon percentage, rather than an assumed value, forms the basis for the CO₂ fixation calculation. The measured carbon mass is converted to its CO₂-equivalent using the molecular weight ratio of CO₂ to elemental carbon ($44 \text{ g mol}^{-1} \div 12 \text{ g mol}^{-1} = 3.67$):

$$\text{CO}_2 \text{ Fixed (g)} = \text{Biomass (g)} \times \text{Carbon Fraction (CHNS)} \times (44/12)$$

Applied to the 60 L reference dataset, this calculation yields a working approximation of:

$$\text{CO}_2 \text{ Fixed (g)} = \text{Biomass (g)} \times 1.833$$

Oxygen evolution is derived from the CHNS-validated CO₂ fixation value using the stoichiometric mass relationship of oxygenic photosynthesis, in which one mole of CO₂ fixed corresponds to approximately one mole of O₂ released (mass ratio 32:44):

$$\text{O}_2 \text{ Evolved (g)} = \text{CO}_2 \text{ Fixed (g)} \times (32/44) = \text{CO}_2 \times 0.727$$

This calibrated equation set — dry weight biomass quantification, CHNS-derived carbon fraction, and stoichiometric CO₂/O₂ conversion — was applied uniformly across the product range, using each unit's culture volume in place of the 60 L reference volume, since all Carbelim systems share the same core microalgal culture and operate at the same validated productivity rate.

4.1.4 Master Equation Set (Volume-Based)

Since biomass productivity, carbon fraction, and the stoichiometric conversion factors are held constant across all Carbelim systems, the relationships derived above can be consolidated into a single standardised equation set, expressed directly as a function of culture volume (V , in litres). This allows performance figures for any Carbelim product to be calculated instantly from its tank volume alone, without repeating the full derivation each time.

Table 3. Empirical Scaling Equations for Biomass Production, Carbon Sequestration, and Oxygen Evolution Across Different Culture Volumes

Parameter	Equation (V = culture volume in litres)	Unit
Biomass production (daily)	$B = 1.8 * V$	g day^{-1}
CO ₂ fixation (daily)	$\text{CO}_2 = 3.30 * V$	g day^{-1}
CO ₂ fixation (annual)	$\text{CO}_2 = 1.2045 * V$	kg year^{-1}
O ₂ evolution (daily)	$\text{O}_2 = 2.40 * V$	g day^{-1}
O ₂ evolution (annual)	$\text{O}_2 = 0.876 * V$	kg year^{-1}

These five equations form the standard reference framework used throughout this report and across Carbelim's technical and commercial documentation. For any product of culture volume V (litres), simply substitute V into the relevant equation to obtain daily or annual biomass, CO₂ fixation, or O₂ evolution figures.

4.2 Results Across the Product Line

Applying these relationships to the biomass figures in Table 2 yields the CO₂ fixation and O₂ evolution figures shown in Table 4.

Table 4. Performance Metrics for CO₂ Sequestration and Oxygen Generation Across Different CARBELIM Models

CO ₂ FIXATION AND O ₂ EVOLUTION PERFORMANCE								
Product	CO ₂				O ₂			
	Daily CO ₂ (g)	Daily CO ₂ (L)	Annual CO ₂ (kg)	Annual CO ₂ (L)	Daily O ₂ (g)	Daily O ₂ (L)	Annual O ₂ (kg)	Annual O ₂ (L)
Miniforest	5.94	30.3	2.17	1106	4.32	3.03	1.58	1107
Air Forest Indoor	211.20	107.58	77.09	39316	153.60	107.58	56.06	39263
Air Forest Outdoor	308.54	157.17	112.62	57414	224.40	157.16	81.90	57358
PRO15	1,633.47	832.04	596.22	303696	1,187.98	831.98	433.61	303669
Wall Mount	90.75	46.24	33.12	16873	66.00	46.23	24.09	16872
Park Bench (Large)	1,501.47	764.79	548.04	279182	1,091.98	764.79	398.57	279143
Park Bench (Small)	465.29	237	169.83	86510	338.39	237	123.51	86497

Note: Gas volumes were calculated from the measured mass of CO₂ fixed and O₂ evolved using the molar volume of an ideal gas at STP (22.414 L mol⁻¹). The conversion factors used were:



1 g CO₂ = 0.5094 L | 1 kg CO₂ = 509.4 L



1 g O₂ = 0.7004 L | 1 kg O₂ = 700.4 L

Accordingly, CO₂ (L) = CO₂ (g) × 0.5094 (or CO₂ (kg) × 509.4) and O₂ (L) = O₂ (g) × 0.7004 (or O₂ (kg) × 700.4). Because photosynthesis consumes one mole of CO₂ and produces one mole of O₂, the resulting gas volumes are approximately equal under the same temperature and pressure conditions, even though their masses differ due to different molecular weights.



Note: The volumetric values of CO₂ fixed and O₂ evolved are approximately equal because oxygenic photosynthesis consumes one mole of CO₂ and produces one mole of O₂. According to Avogadro's law, equal numbers of moles of gases occupy equal volumes under the same temperature and pressure. Consequently, although the mass of CO₂ fixed (44 g mol⁻¹) is greater than the mass of O₂ evolved (32 g mol⁻¹), their gas volumes (L) are essentially identical under standard conditions. Any minor differences in the reported values arise solely from rounding.

The PRO15 leads the range with 1,633.47 g CO₂ fixed per day (596.22 kg/year) and 1,187.98 g O₂ evolved per day (433.61 kg/year) — making it Carbelim's flagship unit for large-scale carbon capture deployments. The Park Bench (Large) follows closely at 548.04 kg CO₂/year. At the entry level, the Miniforest fixes a modest 2.17 kg CO₂/year, appropriate for its compact, personal-use form factor.

6. Ecological Benchmark: Oak Tree Equivalence

To express Carbelim's carbon capture performance in terms accessible to sustainability stakeholders, annual CO₂ fixation has been benchmarked against the published sequestration rate of a mature tree in a restored tropical urban forest. Kumar et al. (2026) report an average annual carbon sequestration of $6.12 \pm 0.39 \text{ kg C tree}^{-1} \text{ year}^{-1}$, equivalent to $22.44 \text{ kg CO}_2 \text{ tree}^{-1} \text{ year}^{-1}$:

$$\text{Tree Equivalent} = \text{Annual CO}_2 \text{ Fixed (kg)} \div 22.44$$

Table 5: Oak Tree Equivalence by Product

Product	Annual CO ₂ Fixed (kg)	Equivalent Mature Oak Trees
Miniforest	2.17	0.1
Air Forest Indoor	77.09	3.4
Air Forest Outdoor	112.62	5.0
PRO15	596.22	26.6
Wall Mount	33.12	1.5
Park Bench (Large)	548.04	24.4
Park Bench (Small)	169.83	7.6

On this basis, the PRO15 (495 L) delivers carbon capture equivalent to approximately 26.6 mature oak trees, while the Park Bench (Large) is equivalent to approximately 24.4 trees. The Air Forest Outdoor and Air Forest Indoor units fall in the mid-range at 5.0 and 3.4 trees, respectively. The Miniforest, given its compact scale, is equivalent to approximately 0.1 trees — illustrating that this entry-level unit is best positioned as a demonstration, educational, or personal-space product rather than a primary carbon offset solution.

It is important to emphasise that, unlike trees, Carbelim PBR units operate continuously regardless of season, require no land area beyond their installation footprint, and can be deployed indoors or in urban settings where tree planting is not feasible. The biomass generated additionally carries downstream valorisation potential (e.g. as feedstock for biofuels, nutraceuticals, or animal feed), a benefit unavailable from standing forest carbon stock.

7. Comparative Performance Assessment and Conclusions

This report establishes a validated, scalable performance framework for the full Carbelim product line, grounded in the experimentally confirmed biomass productivity of $1.8 \text{ g L}^{-1} \text{ day}^{-1}$ measured on the reference 60 L system. Key conclusions:

CO₂ fixation scales linearly with tank volume, ranging from 2.17 kg/year (Miniforest) to 596.22 kg/year (PRO15) across the product range.

Oxygen evolution tracks proportionally with CO₂ fixation, from 1.58 kg/year to 433.61 kg/year across the range.

Benchmarked against mature oak/tropical urban forest trees (Kumar et al., 2026), the product range spans the equivalent of 0.1 to 26.6 trees per unit, offering deployable, space-efficient, and weather-independent carbon capture compared to conventional afforestation.

The product line allows customers to select the appropriately scaled solution — from personal/educational (Miniforest, Wall Mount) to institutional/large-scale (PRO15, Park Bench (Large)) — without compromising on the underlying biological performance per litre of culture.

Overall, the CARBELIM product portfolio demonstrates a scalable and scientifically validated microalgal carbon capture platform, capable of delivering predictable biomass production, CO₂ sequestration, and oxygen generation across a wide range of deployment scales. The quantitative performance framework presented in this report provides a robust basis for product selection, techno-economic evaluation, ESG reporting, and future commercial deployment decisions.

8. References

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9. Annexures



WALL MOUNT



MINI FOREST



PRO 15